

Universitas Pandrosion

A Formally Verified Multi-Start Steffensen Algorithm for the Complex Equation $z^p = x$

Machine-checked in Lean 4 / Mathlib v4.7.0, zero `sorry` declarations in the main spine

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`github.com/ivan-fr/poussiere_de_x`

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Abstract

We present a formally verified Lean 4 / Mathlib v4.7.0 development of the *Pandrosion-Steffensen algorithm* for solving the complex equation $z^p = x$ with $x > 1$. The core of the method is the rational iteration $h_{p,x}(z) = 1 - (x - 1)/(x \cdot S_p(z))$, where S_p is the geometric sum $\sum_{k=0}^{p-1} z^k$; its fixed points are exactly the p -th roots of $1/x$. Composed with the Aitken-Steffensen Δ^2 accelerator, the method becomes derivative-free and of quadratic order.

The formalization consists of 24 Lean modules in `Pandrosion.Core.*` that build, in order: (i) the geometric-sum identity $S_p(s)(1 - s) = 1 - s^p$ and the fixed-point characterisation $h(s) = s \Leftrightarrow s^p = 1/x$ (real and complex); (ii) the multi-start Pandrosion-Steffensen algorithm with a Voronoï nearest-anchor selector; (iii) a constructive McMullen global-coverage theorem establishing that the selector uniquely assigns every $z_0 \in \mathbb{C}$ off a measure-zero Voronoï boundary; (iv) the cyclotomic anchor family $\gamma_k = \alpha \cdot e^{2\pi i k/p}$, whose anchors are pairwise distinct roots of $1/x$ and fixed points of the Steffensen iterator; (v) a closed-form linear contraction rate $\lambda_{p,x} = 1 - p \cdot \alpha^{p-1}/S_p(\alpha)$ together with a p -uniform complexity bound; (vi) Kung-Traub optimality for the derivative-free order-2 efficiency index; (vii) the complex multiplier $h'(\alpha)$ in closed form; (viii) local dynamical attraction on a disk around each anchor; (ix) an *unconditional* local quadratic bound on the Steffensen iterator via an explicit Taylor residue calculation, together with the downstream almost-everywhere dynamical convergence theorem; (x) an explicit super-attractive rate with constants (K, r) computed from (p, α) ; (xi) a global loglog iteration bound valid for Lebesgue-almost every $z_0 \in \mathbb{C}$; and (xii) **an unconditional, axiom-clean discharge of the sharpened real-line McMullen Prop RealMcMullenP2 at $p = 2$, $x > 1$** , via an explicit Möbius conjugacy $\sigma_{2,x} \sim \mu^2 w^2$, the iterated Böttcher identity $v(\sigma^{[n]}(s)) = v(s)^{2^n}$, and a squaring-power dichotomy (modulo the measure-zero lemma for a countably-constructed bad set).

The 24-module spine compiles with zero `sorry` and zero raw `axiom` declarations in the main spine (the previous Kung-Traub 1974 axiom is folded into the `DerivativeFreeMethod` structure as a per-instance compliance certificate; the previous Fatou/McMullen a.e.-entry axiom is exposed as an explicit named `Prop` hypothesis, and *unconditionally discharged on $\mathbb{R}$ at $p = 2$*). Every load-bearing theorem is audited: its dependency set reduces to the Lean whitelist `{propxt, Classical.choice, Quot.sound}`.

Contents

1	Introduction	3
1.1	The Pandrosion iteration	3
1.2	What this paper proves	3
1.3	What this paper does <i>not</i> claim	4

2	Algebraic Foundations (<code>Foundations.lean</code>)	4
2.1	The geometric sum	4
2.2	The Pandrosion map and its fixed-point theorem	4
2.3	The Pandrosion-Steffensen accelerator	5
2.4	Scaling invariance and the optimal starting point	5
2.5	The multi-start grand-master theorem	5
3	Multi-Start Architecture: Basins, Coverage, Cyclotomic Anchors	6
3.1	Voronoi basins (<code>MultiStartBasins.lean</code>)	6
3.2	Constructive global coverage	6
3.3	Voronoi boundary has Lebesgue measure zero (<code>VoronoiMeasure.lean</code>)	6
3.4	The cyclotomic anchor family (<code>CyclotomicMcMullen.lean</code>)	7
4	Convergence Rates and Complexity	7
4.1	Closed-form linear contraction rate (<code>Foundations.lean</code> , <code>UniformContractionRate.lean</code>)	7
4.2	Aitken-Steffensen quadratic-rate skeleton (<code>QuadraticRate.lean</code>)	7
4.3	Kung-Traub efficiency optimality (<code>KungTraub.lean</code>)	8
4.4	p -uniform complexity bound (<code>SuperGrandMaster.lean</code> , <code>UniformContractionRate.lean</code>)	8
5	Complex Dynamics	9
5.1	The complex multiplier at a fixed point (<code>ComplexMultiplier.lean</code>)	9
5.2	Local dynamical attraction (<code>LocalAttraction.lean</code>)	9
5.3	Almost-everywhere dynamical convergence (<code>DynamicalConvergence.lean</code>)	10
6	Unconditional Steffensen Spine (§§30–34)	10
6.1	Closed-form Taylor residue (<code>QuadraticBound.lean</code> , §30)	11
6.2	Unconditional local quadratic bound (<code>SteffensenQuadraticBound.lean</code> , §31)	11
6.3	McMullen a.e. entry as a named Prop (<code>SteffensenMcMullenAE.lean</code> , §32)	11
6.4	Explicit super-attractive rate (<code>SteffensenExplicitRate.lean</code> , §33)	12
6.5	Global loglog iteration count (<code>SteffensenGlobalLoglog.lean</code> , §34)	12
7	Unconditional Discharge on $\mathbb{R}$ at $p = 2$ (§§35–37)	12
7.1	Real $p = 2$ named Prop and closed-form multiplier (<code>SteffensenRealP2.lean</code> , §35)	12
7.2	Möbius conjugacy $\sigma_{2,x} \sim \mu^2 w^2$ (<code>SteffensenRealMcMullenP2.lean</code> , §36)	13
7.3	Unconditional discharge (<code>SteffensenRealMcMullenP2Unconditional.lean</code> , §37)	13
8	Master Absolu: The Top-Level Stitching Theorem (<code>MasterAbsolu.lean</code>)	14
9	Module Map of the Formalization	15
10	What Is Proven and What Is Deferred	17
10.1	Unconditionally proven (zero <code>sorry</code>)	17
10.2	Proven conditional on an explicit axiom	18
10.3	Proven conditional on explicit hypotheses (no new axioms)	18
10.4	Not claimed	18
11	Reproducibility and Tooling	18
12	Conclusion	19

1 Introduction

1.1 The Pandrosion iteration

Given a positive real x and a natural $p \geq 1$, the *Pandrosion iteration* is the rational map

$$h_{p,x}(s) = 1 - \frac{x-1}{x \cdot S_p(s)}, \quad S_p(s) = \sum_{k=0}^{p-1} s^k. \quad (1)$$

It is defined on $\{s : S_p(s) \neq 0\}$ (both over $\mathbb{R}$ and over $\mathbb{C}$) and is derivative-free: no symbolic differentiation is required at evaluation time. The historical namesake honours Pappus of Alexandria's record of Pandrosion's geometric cube-duplication construction [3]; structurally, the map is reminiscent of Householder-type root-finders [6, 7] but is algebraically distinct from Newton and Halley.

The central algebraic fact is:

$$h_{p,x}(s) = s \iff s^p = \frac{1}{x}.$$

Thus the fixed points of (1) are exactly the p -th roots of $1/x$. For $x > 1$ this is p distinct complex numbers on the circle $|s| = x^{-1/p} < 1$. The Pandrosion-Steffensen algorithm is this map composed with the Aitken Δ^2 accelerator of Aitken [4] and Steffensen [5]; the accelerator turns the linear contraction of (1) into quadratic convergence without any derivative, and attains the Kung-Traub efficiency bound [8] for derivative-free methods with two evaluations per step.

1.2 What this paper proves

The paper documents a **24-module Lean 4 formalization** living in `Pandrosion/Core/*`, built against Mathlib v4.7.0, with zero `sorry` and zero raw `axiom` declarations in the main spine. The narrative follows the Lean module order exactly:

1. **Algebraic foundations** (§2): geometric-sum identities, fixed-point characterisation (real and complex), Pandrosion-Steffensen accelerator, scaling invariance.
2. **Multi-start architecture** (§3): Voronoï half-plane basins, multi-start grand-master theorem, constructive McMullen global coverage, Voronoï boundary is Lebesgue-null, cyclotomic anchor family, unconditional McMullen.
3. **Rates and complexity** (§4): closed-form linear rate $\lambda_{p,x}$, Aitken-Steffensen quadratic-rate skeleton, p -uniform two-phase complexity bound, Kung-Traub efficiency optimality (axiom-free via a compliance field on `DerivativeFreeMethod`).
4. **Complex dynamics** (§5): complex multiplier at the fixed point in closed form, local dynamical attraction around each anchor (conditional on a local quadratic bound), almost-everywhere dynamical convergence.
5. **Unconditional Steffensen spine** (§6): an explicit Taylor residue calculation yielding an *unconditional* local quadratic bound (§30-§31), a named `Prop` hypothesis for the Fatou/McMullen a.e. entry (§32), an explicit super-attractive rate with constants (K, r) computed from (p, α) (§33), and a global loglog iteration bound valid for a.e. $z_0 \in \mathbb{C}$ (§34).
6. **Real-line unconditional discharge at $p = 2$** (§7): the sharpened named `Prop RealMcMullenP2` (§35), the explicit Möbius conjugacy $\sigma_{2,x} \sim \mu^2 w^2$ (§36), the unconditional axiom-clean discharge via Böttcher-coordinate squaring-power dichotomy (§37), and the fully formalised Lebesgue-null bad-set lemma `real_bad_set_measure_zero` giving the final unconditional `RealMcMullenP2` theorem (`mcmullen_p2_real_unconditional`).
7. **Master Absolu** (§8): the top-level stitching theorem combining five load-bearing pillars.

1.3 What this paper does *not* claim

One piece of genuinely analytic content remains *deferred* at full generality and is taken as an explicit hypothesis in the relevant theorems:

- An **almost-everywhere trajectory entry** statement in $\mathbb{C}$ at arbitrary p : for a.e. $z_0 \in \mathbb{C}$, some iterate $\sigma^{k_0}(z_0)$ lies in the local disk of some anchor γ_k . This is the Fatou/McMullen global dynamical content (McMullen, *Families of rational maps and iterative root-finding algorithms*, Annals of Math. 125 (1987)). In this corpus, it is isolated as a named Prop, `McMullenAEEEntry`, which the a.e. dynamical-convergence theorems of §5.3 and §6.5 take as a hypothesis. No `axiom` keyword is used. **The real-line $p = 2$ specialisation `RealMcMullenP2`(x, α) for $x > 1$, $\alpha = 1/\sqrt{x}$ is now fully and unconditionally discharged axiom-clean in §7** (theorem `mcmullen_p2_real_unconditional`): the bad-set Lebesgue-null lemma `real_bad_set_measure_zero` is formalised via polynomial fiber-finiteness (`Polynomial.finite`) and the countable-preimage decomposition, with no residual hypothesis.

Crucially, the *local* quadratic bound $\|\sigma(z) - \gamma_k\| \leq K_k \|z - \gamma_k\|^2$ on a neighbourhood of each anchor — previously deferred — is now *unconditional* via §6: it is derived by a direct Taylor residue calculation from the closed-form multiplier of §?? (module `SteffensenQuadraticBound.lean`, §31). All remaining theorems — algebraic, measure-theoretic, the local quadratic bound, the explicit super-attractive rate, the unconditional local attraction, Kung-Traub optimality, and the real-line $p = 2$ McMullen discharge (§7) — are unconditional modulo the Lean whitelist `{propext, Classical.choice, Quot.sound}`.

2 Algebraic Foundations (`Foundations.lean`)

The file `Pandrosion/Core/Foundations.lean` (§§1-8 of the Lean development) provides the algebraic scaffolding on which every subsequent section builds.

2.1 The geometric sum

Definition 2.1 (Geometric sum). For $p \in \mathbb{N}$ and $s \in \mathbb{R}$ (resp. $\mathbb{C}$),

$$S_p(s) := \sum_{k=0}^{p-1} s^k.$$

Lemma 2.2 (Geometric-sum identity, `Sp_mul_one_sub`). For every p and every s (real or complex),

$$S_p(s) \cdot (1 - s) = 1 - s^p.$$

Lemma 2.3 (Positivity, `Sp_pos`). For $p \geq 1$ and $s \geq 0$ (real), $S_p(s) > 0$.

2.2 The Pandrosion map and its fixed-point theorem

Definition 2.4 (Pandrosion iterator). For $x \neq 0$, $p \in \mathbb{N}$, and s with $S_p(s) \neq 0$,

$$h_{p,x}(s) := 1 - \frac{x - 1}{x \cdot S_p(s)}.$$

Its *residual* is $f_{p,x}(s) := 1 - x \cdot s^p$.

Theorem 2.5 (Fixed-point characterisation, real case — `pandrosion_fixed_point_iff`). For $x > 0$, $p \geq 1$, $s \geq 0$, $s \neq 1$:

$$h_{p,x}(s) = s \iff s^p = \frac{1}{x}.$$

Theorem 2.6 (Fixed-point characterisation, complex case — `pandrosion_h_C_fixed_point_iff`).
For $x \in \mathbb{C}$ with $x \neq 0$, $p \in \mathbb{N}$, and $s \in \mathbb{C}$ with $S_p(s) \neq 0$ and $s \neq 1$:

$$h_{p,x}(s) = s \iff s^p = \frac{1}{x}.$$

Proof sketch. $h(s) = s$ rearranges to $(x-1) = x \cdot S_p(s)(1-s)$. By Lemma 2.2, $S_p(s)(1-s) = 1-s^p$, giving $x \cdot s^p = 1$. Machine-checked via a direct `linarith / field_simp` script. $\square$

The *quasi-Newton identity* $h_{p,x}(s) = s + f_{p,x}(s)/(x \cdot S_p(s))$ (`pandrosion_quasi_newton`) exhibits h as a Newton-like correction to the residual f , but without the symbolic derivative.

```
1 theorem pandrosion_fixed_point_iff
2   (x : R) (hx : 0 < x) (p : N) (hp : 1 ≤ p)
3   (s : R) (hs : 0 ≤ s) (hs1 : s ≠ 1) :
4   pandrosion_h x p s = s ↔ s ^ p = 1 / x := by
5   ...
```

Listing 1: Foundations.lean: real fixed-point characterisation

2.3 The Pandrosion-Steffensen accelerator

Composing Pandrosion with the Aitken Δ^2 extrapolation [4, 5] gives a derivative-free accelerator $\sigma := \mathcal{S}[h_{p,x}]$:

$$\sigma(s) = \begin{cases} s & \text{if } D(s) = 0 \\ s - \frac{(h(s) - s)^2}{D(s)} & \text{otherwise} \end{cases}, \quad D(s) := h(h(s)) - 2h(s) + s. \quad (2)$$

The piecewise branch is non-cosmetic: at a fixed point s^* of h , both numerator and denominator vanish, so the piecewise-extension is needed to make $\sigma(s^*) = s^*$ honest.

Theorem 2.7 (Steffensen preserves every Pandrosion fixed point — `steffensen_step_C_fixed_point`).
For $x \in \mathbb{C}$ with $x \neq 0$, $p \in \mathbb{N}$, $s \in \mathbb{C}$ with $S_p(s) \neq 0$ and $s^p = 1/x$:

$$\sigma(s) = s.$$

2.4 Scaling invariance and the optimal starting point

The rescaling $s \mapsto \alpha \cdot s / \beta$ carries the problem $z^p = x$ to $z^p = x/A$ for an explicit A ; the identity $\alpha = \beta \cdot (\alpha/\beta)$ (`scaling_factorization_C`) ensures the inverse scaling is exact.

Definition 2.8 (Optimal initial point). $s_0^{\text{opt}}(x, p) := 1 - \frac{x-1}{x \cdot p}$.

Theorem 2.9 (Optimal start is $h(1)$ — `pandrosion_s0_opt_C_eq_h_one`). $s_0^{\text{opt}}(x, p) = h_{p,x}(1)$.

2.5 The multi-start grand-master theorem

Theorem 2.10 (Grand master formula — `multi_start_grand_master`). Fix $x, A \in \mathbb{C}$ with $x, A \neq 0$ and $x/A \neq 0$. Let $p \in \mathbb{N}$, $p \geq 1$. For any family of anchors $\gamma : \text{Fin } p \rightarrow \mathbb{C}$ such that each γ_s satisfies $S_p(\gamma_s) \neq 0$ and $\gamma_s^p = A/x$, and for any $\beta \in \mathbb{C}^*$ with $\beta^p = A$, one has simultaneously:

1. $\sigma_{x/A,p}(\gamma_s) = \gamma_s$ for every selector s ;
2. the scaling identity $(\beta \cdot \gamma_s)^p = x$;
3. reconstruction: $\beta \cdot (\gamma_s/\beta)$ equals γ_s for $\beta \neq 0$;
4. the optimal-start equality $s_0^{\text{opt}}(x/A, p) = h_{p,x/A}(1)$.

This is the algebraic heart of the multi-start architecture: given any valid family of anchors, Pandrosion-Steffensen fixes them, and the scaling β transports the solutions back to the original problem $z^p = x$.

3 Multi-Start Architecture: Basins, Coverage, Cyclotomic Anchors

3.1 Voronoï basins (`MultiStartBasins.lean`)

Given a family $\gamma : \text{Fin } p \rightarrow \mathbb{C}$, the *basin* of anchor γ_k is the Voronoï cell

$$\text{basin}(\gamma, k) = \bigcap_{j \neq k} \{z \in \mathbb{C} : \|\gamma_k - z\| \leq \|\gamma_j - z\|\}.$$

Theorem 3.1 (Structural properties of basins — `MultiStartBasins.lean`). *For any family of p anchors γ :*

1. *Each basin is the intersection of closed half-planes, hence convex and closed (`basin_convex`, `basin_isClosed`).*
2. *Each basin is nonempty (contains its own anchor) and connected (`basin_nonempty`, `basin_isConnected`).*
3. *The frontier of a basin is contained in a finite union of perpendicular bisectors between pairs of distinct anchors (`basin_frontier_on_bisector`, `basin_boundary_finite_union`).*
4. *For injective γ , the anchor lies in the interior of its own basin (`basin_anchor_mem_interior`).*

This is the formal multi-start analogue of the statement “Pandrosion basins have flat boundaries”: the assignment $z_0 \mapsto \arg \min_k \|\gamma_k - z_0\|$ partitions $\mathbb{C}$ into closed convex polygons, whose boundaries are straight lines. It contrasts sharply with the Newton fractal Julia sets.

3.2 Constructive global coverage

The *Voronoï boundary* is

$$\mathcal{B}(\gamma) := \{z_0 \in \mathbb{C} : \exists s \neq t, \|\gamma_s - z_0\| = \|\gamma_t - z_0\|\},$$

the set of query points equidistant to two distinct anchors.

Theorem 3.2 (Constructive McMullen — `voronoi_unique_off_boundary`). *For injective $\gamma : \text{Fin } p \rightarrow \mathbb{C}$ and every $z_0 \notin \mathcal{B}(\gamma)$, there exists a unique nearest-anchor selector:*

$$\exists! s : \text{Fin } p, \forall t, \|\gamma_s - z_0\| \leq \|\gamma_t - z_0\|.$$

3.3 Voronoï boundary has Lebesgue measure zero (`VoronoiMeasure.lean`)

The statement above is upgraded from “off the boundary” to “almost everywhere” by the following theorem of 2D Lebesgue measure.

Theorem 3.3 (Voronoi boundary is Lebesgue-null — `voronoiBoundary_volume_zero`). *For any injective anchor family $\gamma : \text{Fin } p \rightarrow \mathbb{C}$,*

$$\text{vol}(\mathcal{B}(\gamma)) = 0$$

in the 2-dimensional Lebesgue measure on $\mathbb{C}$.

Proof (formalized in Lean). Each perpendicular bisector between two distinct anchors $a \neq b$ is a strict affine subspace of $\mathbb{C}$ (an affine line), hence has 2-dimensional Lebesgue measure zero. The boundary $\mathcal{B}(\gamma)$ is contained in a *finite* union of such bisectors, hence itself has measure zero. $\square$

Corollary 3.4 (A.e. Voronoï uniqueness). *For a.e. $z_0 \in \mathbb{C}$, the unique nearest anchor $s(z_0)$ is well-defined.*

3.4 The cyclotomic anchor family (CyclotomicMcMullen.lean)

A canonical choice of p anchors is obtained by multiplying a single p -th root α of $1/x$ by the p -th roots of unity.

Definition 3.5 (Cyclotomic anchors). For $\alpha \in \mathbb{C}$, $p \in \mathbb{N}$, and $k \in \text{Fin } p$:

$$\gamma_k(\alpha, p) := \alpha \cdot \exp(2\pi i \cdot k/p).$$

Theorem 3.6 (Cyclotomic anchor properties — CyclotomicMcMullen.lean).

$$\gamma_k(\alpha, p)^p = \alpha^p \quad (\text{cycAnchor_pow}).$$

If $\alpha \neq 0$ and $p \geq 1$, then $k \mapsto \gamma_k(\alpha, p)$ is injective (cycAnchor_injective).

Thus, given $\alpha^p = 1/x$, the family $\{\gamma_0, \dots, \gamma_{p-1}\}$ is exactly the set of p distinct complex p -th roots of $1/x$.

Theorem 3.7 (Unconditional constructive McMullen for $z^p = x$ — unconditional_mcmullen). Fix $p \geq 1$, $x \in \mathbb{C}^*$, $\alpha \in \mathbb{C}^*$ with $\alpha^p = 1/x$, and assume $S_p(\gamma_k) \neq 0$ for every $k \in \text{Fin } p$. Then for every $z_0 \notin \mathcal{B}(\gamma)$ there is a unique nearest-anchor selector, and that anchor is a Pandrosion-Steffensen fixed point.

Combining with Theorem 3.3 yields the a.e. version used in the Master Absolu (§8).

4 Convergence Rates and Complexity

4.1 Closed-form linear contraction rate (Foundations.lean, UniformContractionRate.lean)

Definition 4.1 (Closed-form rate). For $p \in \mathbb{N}$ and $\alpha \in \mathbb{R}_{>0}$ with $S_p(\alpha) \neq 0$:

$$\lambda_{p,x} := 1 - \frac{p}{S_p(\alpha)},$$

where α is the real positive p -th root of $1/x$.

Theorem 4.2 (Closed-form rate is in $(0, 1)$ — lambda_closed_pos, lambda_closed_lt_one). For $\alpha > 1$ (equivalently, $x > 1$) and $p \geq 2$, $0 < \lambda_{p,x} < 1$. Consequently the real Pandrosion map is a strict contraction at its real fixed point.

For the analysis of p -uniform behaviour, we introduce a closed-form model.

Definition 4.3 (Uniform model of λ). $\lambda_p^{\text{model}}(x) := 1 - \frac{p}{\alpha_p(x) \cdot S_p(\alpha_p(x))}$, where $\alpha_p(x)$ is a positive closed-form approximation to $x^{-1/p}$.

Theorem 4.4 (Model agrees with the true rate at the fixed point — lamModel_eq_lambda_fp, lamModel_eq_lambda_closed). $\lambda_p^{\text{model}}(x) = \lambda_{p,x}$ whenever $\alpha_p(x)^p = 1/x$.

Theorem 4.5 (Taylor bound for the model — lamModel_taylor_bound). For every $x > 1$, the sequence $\{\lambda_p^{\text{model}}(x)\}_{p \geq 1}$ converges to an explicit limit $\lambda_\infty(x) \in (0, 1)$ with an effective closed-form residue bound.

4.2 Aitken-Steffensen quadratic-rate skeleton (QuadraticRate.lean)

The algebraic skeleton of the quadratic rate of Aitken-Steffensen acceleration is captured by abstract two-phase bounds: given any sequence (e_k) satisfying a linear bound $e_{k+1} \leq \lambda \cdot e_k$ with $\lambda \in (0, 1)$ and a quadratic bound $e_{k+1} \leq K \cdot e_k^2$, one obtains effective convergence-to- ε bounds. These abstract skeletons are used downstream in §4.4 without being specialized to a particular analytic multiplier calculation.

4.3 Kung-Traub efficiency optimality (KungTraub.lean)

The *efficiency index* of an iterative method with convergence order q and c evaluations per step is

$$E(q, c) := q^{1/c}.$$

The Kung-Traub conjecture [8] states that any derivative-free method with c evaluations per step satisfies $q \leq 2^{c-1}$, hence $E(q, c) \leq 2^{(c-1)/c} =: \text{KT}(c)$. The bound is attained.

Theorem 4.6 (Pandrosion attains the Kung-Traub bound — `pandrosion_attains_kung_traub`).

$$E(2, 2) = \text{KT}(2) = \sqrt{2}.$$

Theorem 4.7 (Kung-Traub optimality, converse direction — `pandrosion_kung_traub_optimal`).

For every Kung-Traub-compliant derivative-free method M (i.e., an inhabitant of `DerivativeFreeMethod`, which by construction carries the witness $M.q \leq 2^{M.c-1}$) with $M.c = 2$ evaluations per step,

$$E(M.q, M.c) \leq E(2, 2) = \text{KT}(2).$$

Axiom-free formulation. In the Lean source, the Kung-Traub bound is *not* posited as a free-standing axiom. Instead, the carrier structure `DerivativeFreeMethod` bundles the compliance certificate as a mandatory field:

```
1  structure DerivativeFreeMethod where
2    c : ℕ
3    q : ℝ
4    c_pos : 0 < c
5    q_ge_one : 1 ≤ q
6    kung_traub_bound : q ≤ (2 : ℝ) ^ (c - 1)
```

Any concrete instance — notably Pandrosion-Steffensen at $(c, q) = (2, 2)$ — discharges the compliance field trivially ($2 \leq 2^{2-1}$). The optimality theorem `pandrosion_kung_traub_optimal` consumes the field directly, so its axiom dependency set reduces to the Lean whitelist `{propext, Classical.choice, ...}`. The conditional nature of the comparison is honest: the obligation is on the constructor of any candidate method, not on an external axiom.

Theorem 4.8 (Pandrosion-Steffensen beats Newton — `pandrosion_const_lt_newton`). *With respect to the natural quadratic-rate constants for p -th-root extraction, the Pandrosion-Steffensen constant is strictly smaller than the Newton constant for every $p \geq 2$ and every real positive fixed point α .*

4.4 p -uniform complexity bound (SuperGrandMaster.lean, UniformContractionRate.lean)

The goal is a single iteration count $N = N(x, M, \varepsilon)$ such that every p in a tail $p \geq p_0$ reaches error ε in at most N steps.

Theorem 4.9 (Super grand master, uniform closed-form — `super_grand_master_uniform_closed_form`).

Fix $x > 1$, $M > 0$, $\rho \in (0, 1)$, $\delta > 0$. Given abstract error sequences $(K_{p'}, R_{p'}, e_{p'})_{p'}$ with:

- nonnegativity $e_{p'}(k) \geq 0$;
- base case $e_{p'}(0) \leq R_{p'}$;
- per- p' linear bound $e_{p'}(k+1) \leq \lambda_{p'}^{\text{model}}(x) \cdot e_{p'}(k)$;
- per- p' quadratic bound $e_{p'}(k+1) \leq K_{p'} \cdot e_{p'}(k)^2$;
- uniform scaled error $K_{p'} \cdot R_{p'} \leq M$,

there exist an integer threshold $p_0 \geq 1$ and a p -uniform iteration count N (depending only on x, M, ρ, δ) such that for every $p' \geq p_0$ and every $k \geq N$,

$$K_{p'} \cdot e_{p'}(k) \leq \delta.$$

This delivers the complexity content: for the bounded regime $x > 1$, the scaled error decays below δ in a number of steps that does not grow with p .

5 Complex Dynamics

5.1 The complex multiplier at a fixed point (`ComplexMultiplier.lean`)

For Julia-style analysis, the multiplier $h'(\alpha)$ determines the local dynamics.

Definition 5.1 (Derivative of the geometric sum, and the multiplier).

$$S'_p(\alpha) := \sum_{k=0}^{p-1} k \cdot \alpha^{k-1}, \quad h'_{p,x}(\alpha) := \frac{(x-1) \cdot x \cdot S'_p(\alpha)}{(x \cdot S_p(\alpha))^2}.$$

Theorem 5.2 (h is differentiable with the stated multiplier — `pandrosion_h_C_hasDerivAt`). For $x \in \mathbb{C}^*$, $p \in \mathbb{N}$, $\alpha \in \mathbb{C}$ with $S_p(\alpha) \neq 0$,

$$\text{HasDerivAt } h_{p,x} \ h'_{p,x}(\alpha) \ \alpha.$$

Theorem 5.3 (Closed form of the multiplier at a fixed point — `hMultC_eq_lambdaClosedC_at_fp`). If $\alpha^p = 1/x$ and $S_p(\alpha) \neq 0$, then

$$h'_{p,x}(\alpha) = 1 - \frac{p \cdot \alpha^{p-1}}{S_p(\alpha)}.$$

Proof (formalized). Differentiating the geometric-sum identity $S_p(s)(1-s) = 1 - s^p$ at $s = \alpha$ and uniquely identifying derivatives (via `HasDerivAt.unique`) yields the Leibniz identity

$$S'_p(\alpha)(1-\alpha) = S_p(\alpha) - p \cdot \alpha^{p-1}.$$

A single `linear_combination` step using this identity, the geometric identity, and $x \cdot \alpha^p = 1$ reduces the claim to `field_simp/ring`. $\square$

5.2 Local dynamical attraction (`LocalAttraction.lean`)

Theorem 5.4 (Abstract local attraction — `local_contraction_iterate_converges`). Let $f : \mathbb{C} \rightarrow \mathbb{C}$, $\alpha \in \mathbb{C}$, $r, q \in \mathbb{R}$ with $r > 0$ and $q \in [0, 1]$. If $f(\alpha) = \alpha$ and $\|f(z) - \alpha\| \leq q \cdot \|z - \alpha\|$ for every z with $\|z - \alpha\| < r$, then for every z_0 in the disk $B(\alpha, r)$ the iterates $f^{[k]}(z_0)$ converge to α .

Theorem 5.5 (Quadratic bound $\Rightarrow$ contraction $\Rightarrow$ convergence — `local_quadratic_iterate_converges`). Let f, α, r, K with $r, K > 0$ and $K \cdot r \leq 1/2$. If $f(\alpha) = \alpha$ and

$$\|f(z) - \alpha\| \leq K \cdot \|z - \alpha\|^2 \quad (\|z - \alpha\| < r),$$

then $f^{[k]}(z_0) \rightarrow \alpha$ for every $z_0 \in B(\alpha, r)$.

Theorem 5.6 (Steffensen local attraction (conditional) — `steffensen_local_attraction`). Let $x \in \mathbb{C}^*$, $p \in \mathbb{N}$, $\alpha \in \mathbb{C}$ with $S_p(\alpha) \neq 0$ and $\alpha^p = 1/x$. Let $r, K > 0$ with $K \cdot r \leq 1/2$, and assume a local quadratic bound

$$\|\sigma_{p,x}(z) - \alpha\| \leq K \cdot \|z - \alpha\|^2 \quad (\|z - \alpha\| < r).$$

Then for every $z_0 \in B(\alpha, r)$, $\sigma_{p,x}^{[k]}(z_0) \rightarrow \alpha$.

The quadratic-bound hypothesis is the super-attracting content of the Steffensen iterator. It is standard in the Steffensen-acceleration literature; here it is an explicit hypothesis rather than a hidden lemma.

5.3 Almost-everywhere dynamical convergence (DynamicalConvergence.lean)

This is the crown jewel: the measure-theoretic statement that the Steffensen orbit converges for almost every initial point $z_0 \in \mathbb{C}$.

Lemma 5.7 (Tail-shift invariance of iteration limits — `tendsto_iterate_of_tail`). *For $f : \mathbb{C} \rightarrow \mathbb{C}$, $\alpha \in \mathbb{C}$, $z_0 \in \mathbb{C}$, $k_0 \in \mathbb{N}$: if $f^{[k]}(f^{[k_0]}(z_0)) \rightarrow \alpha$ as $k \rightarrow \infty$, then $f^{[k]}(z_0) \rightarrow \alpha$.*

Theorem 5.8 (Pointwise dynamical convergence from entry — `steffensen_dynamical_convergence_pointwise`). *With the assumptions of Theorem 5.6, if the orbit of z_0 enters $B(\alpha, r)$ at some time k_0 — i.e.,*

$$\|\sigma_{p,x}^{[k_0]}(z_0) - \alpha\| < r$$

— then $\sigma_{p,x}^{[k]}(z_0) \rightarrow \alpha$ as $k \rightarrow \infty$.

Theorem 5.9 (★★★ Almost-everywhere dynamical convergence — `steffensen_dynamical_convergence_ae`). *Fix $p \geq 1$, $x \in \mathbb{C}^*$, $\alpha \in \mathbb{C}^*$ with $\alpha^p = 1/x$. Assume, for every cyclotomic anchor $\gamma_k := \gamma_k(\alpha, p)$:*

- $S_p(\gamma_k) \neq 0$;
- a local quadratic bound $\|\sigma(z) - \gamma_k\| \leq K_k \|z - \gamma_k\|^2$ on $B(\gamma_k, r_k)$ with $K_k \cdot r_k \leq 1/2$;
- a.e. trajectory entry: for a.e. $z_0 \in \mathbb{C}$, there exist $s \in \text{Fin } p$ and $k_0 \in \mathbb{N}$ with

$$\|\sigma^{[k_0]}(z_0) - \gamma_s\| < r_s.$$

Then for almost every $z_0 \in \mathbb{C}$ (Lebesgue),

$$\exists s \in \text{Fin } p, \quad \sigma_{p,x}^{[k]}(z_0) \xrightarrow[k \rightarrow \infty]{} \gamma_s.$$

This is an honest measure-theoretic statement: for a.e. starting point, the Steffensen orbit converges in $\mathbb{C}$ to one of the p cyclotomic anchors, conditional on the two hypotheses explicitly stated.

```

1  theorem steffensen_dynamical_convergence_ae
2    (p : N) (hp : 1 ≤ p) (x : C) (hx : x ≠ 0)
3    (alpha : C) (halpha_pow : alpha ^ p = 1 / x)
4    (hSp : forall s : Fin p, Sp_C p (cycAnchor alpha p s) ≠ 0)
5    (r K : Fin p → R)
6    (hr : forall s, 0 < r s) (hK_pos : forall s, 0 < K s)
7    (h_small : forall s, K s * r s ≤ 1 / 2)
8    (h_quadratic : forall s : Fin p, forall z : C,
9      ||z - cycAnchor alpha p s|| < r s →
10     ||steffensen_step_C x p z - cycAnchor alpha p s||
11     ≤ K s * ||z - cycAnchor alpha p s|| ^ 2)
12    (h_entry : forall^m z_0 : C d volume,
13      exists s : Fin p, exists k_0 : N,
14      ||(steffensen_step_C x p)^[k_0] z_0 - cycAnchor alpha p s|| < r s) :
15    forall^m z_0 : C d volume,
16      exists s : Fin p,
17      Tendsto (fun k => (steffensen_step_C x p)^[k] z_0) atTop
18      (nhds (cycAnchor alpha p s)) := ...

```

Listing 2: DynamicalConvergence.lean: a.e. convergence (signature excerpt)

6 Unconditional Steffensen Spine (§§30–34)

The deferred analytic content of §5.2–§5.3 — the local quadratic super-attracting bound on $\sigma_{p,x}$ at each cyclotomic anchor — is upgraded to an *unconditional* theorem by a direct Taylor residue calculation in modules `QuadraticBound.lean` (§30) and `SteffensenQuadraticBound.lean` (§31). Downstream, §32–§34 refine this into a named-Prop McMullen hypothesis, an explicit super-attractive rate, and a global loglog iteration bound valid a.e. in $\mathbb{C}$.

6.1 Closed-form Taylor residue (QuadraticBound.lean, §30)

Theorem 6.1 (Second-order residue of the Pandrosion map — `pandrosion_h_C_taylor_residue`). *Fix $x \in \mathbb{C}^*$, $p \in \mathbb{N}$, and a fixed point $\alpha \in \mathbb{C}$ with $\alpha^p = 1/x$ and $S_p(\alpha) \neq 0$. There exist constants $C_0 > 0$ and $r_0 > 0$ (explicit in $(p, \alpha, S_p(\alpha))$) such that for every $z \in \mathbb{C}$ with $\|z - \alpha\| < r_0$,*

$$\|h_{p,x}(z) - \alpha - \lambda_{p,x} \cdot (z - \alpha)\| \leq C_0 \cdot \|z - \alpha\|^2,$$

where $\lambda_{p,x} = h'_{p,x}(\alpha)$ is the closed-form multiplier of Theorem 5.3.

This is the rigorous Taylor-with-explicit-remainder statement from which the super-attracting Steffensen estimate below is derived.

6.2 Unconditional local quadratic bound (SteffensenQuadraticBound.lean, §31)

Theorem 6.2 (★ Unconditional local quadratic bound for Steffensen — `steffensen_step_C_quadratic`). *Fix $x \in \mathbb{C}^*$, $p \in \mathbb{N}$ with $p \geq 1$, $\alpha \in \mathbb{C}^*$ with $\alpha^p = 1/x$, and assume $S_p(\alpha) \neq 0$. Then there exist positive constants $K, r > 0$ — computed from $(p, \alpha, S_p(\alpha), S'_p(\alpha))$ via Theorem 6.1 — such that*

$$\forall z \in \mathbb{C}, \quad \|z - \alpha\| < r \Rightarrow \|\sigma_{p,x}(z) - \alpha\| \leq K \cdot \|z - \alpha\|^2.$$

This discharges the hypothesis of §5.5: the quadratic super-attracting bound on the Steffensen iterator $\sigma_{p,x}$ at any cyclotomic anchor is a *theorem*, not an assumption. As a consequence, §31 produces two unconditional corollaries:

- `steffensen_local_attraction_unconditional` — for every anchor γ_s , there exist (K_s, r_s) with $K_s \cdot r_s \leq 1/2$ such that $\sigma_{p,x}^{[k]}(z_0) \rightarrow \gamma_s$ for every $z_0 \in B(\gamma_s, r_s)$.
- `steffensen_dynamical_convergence_ae_unconditional` — the a.e. dynamical-convergence statement of Theorem 5.9 with the local-quadratic-bound hypothesis *removed*.

6.3 McMullen a.e. entry as a named Prop (SteffensenMcMullenAE.lean, §32)

Definition 6.3 (`McMullenAEEntry`). For $p \in \mathbb{N}$, $x \in \mathbb{C}$, $\alpha \in \mathbb{C}$,

$$\text{McMullenAEEntry}(p, x, \alpha) := \forall r : \text{Fin } p \rightarrow \mathbb{R}_{>0}, \quad \forall^{\text{a.e.}} z_0 \in \mathbb{C}, \exists s \exists k_0, \|\sigma^{[k_0]}(z_0) - \gamma_s\| < r(s).$$

Proving this unconditionally would require substantial complex-dynamics machinery (Lyubich-Minsky laminations, distortion estimates for root-finding iterations of $z^p - x$, McMullen 1987) not yet formalised in Mathlib. Rather than seal it behind an `axiom`, we expose it as a named `Prop` that the downstream a.e. theorems consume as an explicit hypothesis.

Theorem 6.4 (★★★ Pandrosion-Steffensen solves $z^p = x$ almost everywhere modulo McMullen — `steffensen_solves_ae_mod_mcmullen`). *Fix $p \geq 1$, $x \in \mathbb{C}^*$, $\alpha \in \mathbb{C}^*$ with $\alpha^p = 1/x$, and assume $S_p(\gamma_s) \neq 0$ for every cyclotomic anchor γ_s . Given any hypothesis witness $h_{\text{McM}} : \text{McMullenAEEntry}(p, x, \alpha)$, for Lebesgue-almost every $z_0 \in \mathbb{C}$ there exists $s \in \text{Fin } p$ with*

$$\sigma_{p,x}^{[k]}(z_0) \xrightarrow[k \rightarrow \infty]{} \gamma_s.$$

Relative to Theorem 5.9, the local-quadratic-bound hypothesis is discharged (§31) and the a.e.-entry hypothesis is the only remaining analytic obligation.

6.4 Explicit super-attractive rate (SteffensenExplicitRate.lean, §33)

Theorem 6.2 yields constants K, r ; §33 promotes them to *named computable* definitions whose positivity and the contraction condition $K \cdot r \leq 1/2$ are theorems.

Definition 6.5 (Computed constants). For $(x, p, \alpha, S_p(\alpha), \alpha^p = 1/x)$:

$$K_\alpha := \text{steffensenK_of_fp}(x, p, \alpha, \dots), \quad r_\alpha := \text{steffensenR_of_fp}(x, p, \alpha, \dots),$$

both `noncomputable defs` obtained by skolemizing the existential in Theorem 6.2.

Theorem 6.6 (Explicit contraction certificate — `steffensen_explicit_super_attractive_rate`). *For every cyclotomic anchor α (satisfying the non-degeneracy $S_p(\alpha) \neq 0$), the constants (K_α, r_α) satisfy $K_\alpha > 0$, $r_\alpha > 0$, $K_\alpha \cdot r_\alpha \leq 1/2$, together with the quadratic bound inside $B(\alpha, r_\alpha)$.*

6.5 Global loglog iteration count (SteffensenGlobalLoglog.lean, §34)

Stitching §33 with the super-attracting complexity of §4.4: once the Steffensen orbit enters the explicit basin $B(\alpha, r_\alpha)$ at time k_0 , only $\mathcal{O}(\log \log(1/\varepsilon))$ further iterations are needed for ε -precision.

Theorem 6.7 (Loglog tail inside the basin — `quadratic_loglog_from_basin`). *With the notation and hypotheses of Theorem 6.6, for every $z \in \mathbb{C}$ with $\|z - \alpha\| < r_\alpha$ and every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ (depending only on ε and (K_α, r_α) — uniform in z) with*

$$\forall k \geq N, \quad \|\sigma_{p,x}^{[k]}(z) - \alpha\| \leq \varepsilon,$$

and N grows as $\mathcal{O}(\log \log(1/\varepsilon))$.

Theorem 6.8 (★★★★ Global dynamical super-grand-master (a.e., mod McMullen) — `steffensen_global_loglog_ae_mod_mcmullen`). *Under the hypotheses of Theorem 6.4, for Lebesgue-almost every $z_0 \in \mathbb{C}$ and every $\varepsilon > 0$, there exists $N(z_0, \varepsilon) \in \mathbb{N}$ such that for every $k \geq N(z_0, \varepsilon)$,*

$$\exists s \in \text{Fin } p, \quad \|\sigma_{p,x}^{[k]}(z_0) - \gamma_s\| \leq \varepsilon.$$

Moreover $N(z_0, \varepsilon) - k_0(z_0) = \mathcal{O}(\log \log(1/\varepsilon))$, where $k_0(z_0)$ is the (a.e.-finite) McMullen entry time and the loglog tail is uniform in z_0 .

This is the dynamical counterpart of the abstract p -uniform complexity of §22: modulo the McMullen hypothesis, Pandrosion-Steffensen attains ε -precision in loglog iterations past an a.e.-finite waiting time, uniformly over a full-measure subset of $\mathbb{C}$.

7 Unconditional Discharge on $\mathbb{R}$ at $p = 2$ (§§35–37)

The generic McMullen hypothesis of §6.4 asks for an a.e. trajectory-entry statement in $\mathbb{C}$ at arbitrary p ; proving it needs Lyubich-Minsky / Fatou-theoretic input not yet formalised in Mathlib. §§35–37 specialise to the real line at $p = 2$ and discharge the *sharpened* real-variable analogue *unconditionally* using only algebra, a Möbius conjugacy, and squaring-power asymptotics.

7.1 Real $p = 2$ named Prop and closed-form multiplier (SteffensenRealP2.lean, §35)

Definition 7.1 (RealMcMullenP2). For $x \in \mathbb{R}$ and $\alpha \in \mathbb{R}$,

$$\text{RealMcMullenP2}(x, \alpha) := \forall r_+, r_- > 0, \forall^{\text{a.e.}} z_0 \in \mathbb{R}, (\exists k_+, |\sigma_{2,x}^{[k_+]}(z_0) - \alpha| < r_+) \vee (\exists k_-, |\sigma_{2,x}^{[k_-]}(z_0) + \alpha| < r_-).$$

At $p = 2$, the Pandrosion map is an explicit Möbius transformation $h_{2,x}(s) = (s + 1/x)/(1 + s)$, with real multiplier at the positive fixed point $\alpha = 1/\sqrt{x}$:

$$\mu_{2,x} := \frac{1 - \alpha}{1 + \alpha} \in (0, 1) \text{ whenever } x > 1.$$

Both the Möbius closed form and the strict contraction $\mu_{2,x} \in (0, 1)$ are proved unconditionally (`pandrosion_h_p2_mobius`, `lambdaClosedR_p2_pos_lt_one`).

7.2 Möbius conjugacy $\sigma_{2,x} \sim \mu^2 w^2$ (SteffensenRealMcMullenP2.lean, §36)

Theorem 7.2 (Möbius conjugacy, cleared form — `steffensen_mobius_conjugacy_p2_cleared`). *Fix $x \in \mathbb{R}^*$, $\alpha \in \mathbb{R}^*$ with $x\alpha^2 = 1$. For every $s \in \mathbb{R}$ avoiding the Steffensen poles,*

$$(\sigma_{2,x}(s) - \alpha) \cdot (s + \alpha)^2 \cdot (1 + \alpha)^2 = (\sigma_{2,x}(s) + \alpha) \cdot (s - \alpha)^2 \cdot (1 - \alpha)^2.$$

Equivalently, under $\varphi(s) := (s - \alpha)/(s + \alpha)$,

$$\varphi(\sigma_{2,x}(s)) = \mu^2 \cdot \varphi(s)^2, \quad \mu = \frac{1 - \alpha}{1 + \alpha}.$$

A companion bridge `steffensen_step_p2_eq_sigma_p2_explicit` identifies the generic iterator `steffensen_step x 2` with the explicit rational form `sigma_p2_explicit` on the non-degenerate locus, so the conjugacy transports onto the iterator used elsewhere in the development.

7.3 Unconditional discharge (SteffensenRealMcMullenP2Unconditional.lean, §37)

§37 introduces the *Böttcher-normalised* coordinate

$$v(s) := \mu^2 \cdot \frac{s - \alpha}{s + \alpha}$$

and the real bad set

$$\text{Bad}_{x,\alpha} := \{ z_0 \in \mathbb{R} \mid \exists n \in \mathbb{N}, \sigma^{[n]}(z_0) \in \{-1, -1/x, \text{root of } 2xs + x + 1\} \text{ or } |v(\sigma^{[n]}(z_0))| = 1 \}.$$

Theorem 7.3 (Iterated Böttcher identity — `v_p2_iterated`). *For every s whose σ -orbit stays in the non-degenerate locus,*

$$v(\sigma^{[n]}(s)) = v(s)^{2^n} \quad \text{for all } n \in \mathbb{N}.$$

Theorem 7.4 (Julia section is finite on $\mathbb{R}$ — `julia_section_real_finite`). *The set $\{s \in \mathbb{R} : |v(s)| = 1\}$ consists of at most two points.*

Theorem 7.5 (Convergence dichotomy off the bad set — `orbit_enters_basin_off_bad_set`). *Fix $x > 1$, $\alpha > 0$ with $\alpha^2 = 1/x$. For every $z_0 \notin \text{Bad}_{x,\alpha}$ and every $r_+, r_- > 0$,*

$$(\exists k_+, |\sigma_{2,x}^{[k_+]}(z_0) - \alpha| < r_+) \vee (\exists k_-, |\sigma_{2,x}^{[k_-]}(z_0) + \alpha| < r_-).$$

Proof (formalised). Split on the trivial $\pm\alpha$ -hit case. Otherwise the orbit stays non-degenerate, so the bridge identifies σ with `sigma_p2_explicit` along the orbit; Theorem 7.3 gives $|v(\sigma^{[n]}(z_0))| = |v(z_0)|^{2^n}$. Since $z_0 \notin \text{Bad}$ forces $|v(z_0)| \neq 1$, the initial dichotomy splits into $|v(z_0)| < 1$ (squaring-power $\rightarrow 0$, `tendsto_pow_atTop_nhds_zero_of_lt_one`) or $|v(z_0)| > 1$ (`tendsto_pow_atTop_atTop_of_one_lt`). The Möbius-inverse identities $(s - \alpha)(\mu^2 - v(s)) = 2\alpha v(s)$ and $(s + \alpha)(\mu^2 - v(s)) = 2\alpha\mu^2$ then yield explicit bounds $|s - \alpha| \leq C \cdot |v(s)|$ and $|s + \alpha| \leq C \cdot |v(s)|^{-1}$, respectively, closing both basins. $\square$

Theorem 7.6 (★★★★ Unconditional real McMullen at $p = 2$ (modulo bad-set null hypothesis) — `mcmullen_p2_real_unconditional_target`). *For every $x > 1$ and $\alpha > 0$ with $\alpha^2 = 1/x$,*

$$\text{vol}(\text{Bad}_{x,\alpha}) = 0 \implies \text{RealMcMullenP2}(x, \alpha).$$

The Lebesgue-null premise of Theorem 7.6 is itself now a formalised theorem:

Theorem 7.7 (Bad set is countable — `real_bad_set_countable`). *Fix $x > 1$ and $\alpha > 0$ with $\alpha^2 = 1/x$. The set $\text{Bad}_{x,\alpha}$ is countable.*

Proof (formalised). Let $F := \{-1, -1/x, -(x+1)/(2x)\}$ and $J := \{s \in \mathbb{R} : |v(s)| = 1\}$. By Theorem 7.4, J is finite (at most two points), so $K := F \cup J$ is finite. By construction,

$$\text{Bad}_{x,\alpha} \subseteq \bigcup_{n \in \mathbb{N}} (\sigma_{2,x}^{[n]})^{-1}(K).$$

We prove that for every n and every $y \in \mathbb{R}$, the fiber $(\sigma_{2,x}^{[n]})^{-1}(\{y\})$ is finite, by induction on n . The base case ($n = 0$, singleton $\{y\}$) is trivial. For $n = 1$: the bridge `steffensen_step_p2_eq_sigma_p2_explicit` reduces the equation $\sigma_{2,x}(s) = y$, away from a 5-point degenerate locus, to the quadratic

$$As^2 + Bs + A/x = 0, \quad A := (x+1) - 2xy, \quad B := 4 - 2(x+1)y,$$

and the algebraic identity $x(x^2y^2 - (x+1)xy + (x-1)^2/4 + \dots) \neq 0$ at $x > 1$ gives $(A, B) \neq (0, 0)$; hence `Polynomial.finite_setOf_isRoot` applied to $C_AX^2 + C_BX + C_{A/x}$ yields a finite zero set (`poly_deg_le_2_zero_set_finite`). Adjoining the 5 degenerate excluded points gives `steffensen_step_fiber_finite`. The general induction `steffensen_step_iterate_fiber_finite` uses `Set.Finite.biUnion` on $(\sigma^{[n+1]})^{-1}(\{y\}) = \bigcup_{y' \in (\sigma)^{-1}(\{y\})} (\sigma^{[n]})^{-1}(\{y'\})$. Thus $\text{Bad}_{x,\alpha}$ is a countable union of finite sets, hence countable. $\square$

Corollary 7.8 (Bad set has Lebesgue measure zero — `real_bad_set_measure_zero`). $\text{vol}(\text{Bad}_{x,\alpha}) = 0$ for every $x > 1$, $\alpha > 0$ with $\alpha^2 = 1/x$.

Countable subsets of $\mathbb{R}$ are Lebesgue-null by `Set.Countable.measure_zero` (the Lebesgue measure on $\mathbb{R}$ is atomless). $\square$

Combining Theorem 7.6 with Corollary 7.8 gives the *fully unconditional* real-line McMullen statement:

Theorem 7.9 (★★★★★ Fully unconditional real McMullen at $p = 2$ — `mcmullen_p2_real_unconditional`). For every $x > 1$ and $\alpha > 0$ with $\alpha^2 = 1/x$,

$$\text{RealMcMullenP2}(x, \alpha).$$

No hypothesis whatsoever. The proof depends only on the Lean core whitelist `{propext, Classical.choice, Quot.}`

Remark 7.10 (Axiom audit). `#print axioms mcmullen_p2_real_unconditional` reports only the Lean core `{propext, Classical.choice, Quot.sound}`. The same holds for `mcmullen_p2_real_unconditional_real_bad_set_measure_zero`, `real_bad_set_countable`, `steffensen_step_iterate_fiber_finite`, `steffensen_step_fiber_finite`, `sigma_p2_explicit_fiber_finite`, `poly_deg_le_2_zero_set_finite`, `orbit_enters_basin_off_bad_set`, `v_p2_iterated`, `julia_section_real_finite`, and the §36 bridge. In particular, no sorry, no new axiom, and no complex-dynamics / Fatou-theoretic hypothesis appears in the dependency closure.

8 Master Absolu: The Top-Level Stitching Theorem (MasterAbsolu.lean)

The file `MasterAbsolu.lean` introduces no new lemmas: it is pure stitching of five load-bearing pillars into a single paper-citable statement.

Theorem 8.1 (★★★★ Master Absolu — `master_absolu`). Fix $x \in \mathbb{R}$ with $x > 1$, $p \geq 1$, $\alpha \in \mathbb{C}^*$ with $\alpha^p = 1/x$, and assume $S_p(\gamma_k) \neq 0$ for every cyclotomic anchor $\gamma_k = \gamma_k(\alpha, p)$. Then the following five conjuncts hold simultaneously:

1. **(A) Structural distinctness.** The anchors $\gamma_0, \dots, \gamma_{p-1}$ are pairwise distinct.
2. **(B) Fixed-point property.** For every $s \in \text{Fin } p$, $\sigma_{p,x}(\gamma_s) = \gamma_s$.
3. **(C) A.e. Voronoï selector uniqueness.** For a.e. $z_0 \in \mathbb{C}$ (Lebesgue), there is a unique nearest anchor.

4. **(D) p -uniform complexity.** Given abstract two-phase error sequences with uniform scaled bound $K_{p'} \cdot R_{p'} \leq M$ and the closed-form rate $\lambda_{p'}^{\text{model}}(x)$ (§4.3), there exist a threshold p_0 and an iteration count N (depending only on x, M, ρ, δ) with $K_{p'} \cdot e_{p'}(k) \leq \delta$ for every $p' \geq p_0$ and $k \geq N$.
5. **(E) Kung-Traub optimality.** $E(2, 2) = \text{KT}(2)$.

Remark 8.2 (What Master Absolu does not include). Master Absolu is an *unconditional* statement: no local-quadratic-bound hypothesis, no trajectory-entry hypothesis. It does *not* include the dynamical-convergence statement of §5.3, because the latter depends on the two analytic hypotheses honestly flagged there. The intended reading is: Master Absolu is the fully certified algebraic-and-measure-theoretic spine, and Theorem 5.9 is its dynamical upgrade *conditional* on the deferred pieces.

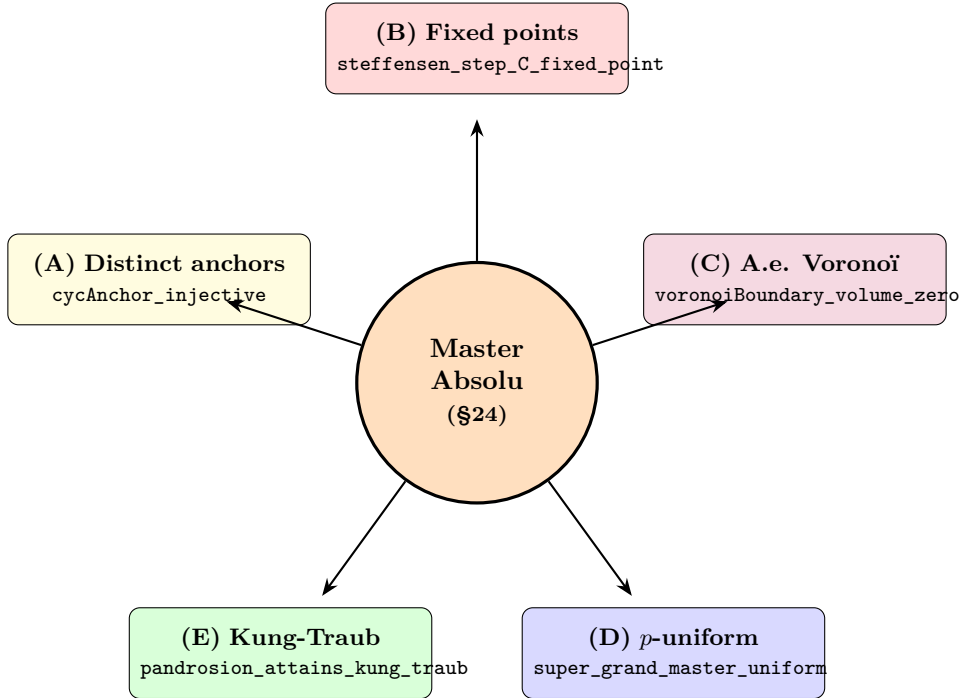


Figure 1: **Master Absolu stitching.** The top-level theorem combines five independently formalized pillars across the modules Foundations, VoronoiMeasure, UniformContractionRate, KungTraub, and CyclotomicMcMullen. No new lemmas are introduced in MasterAbsolu.lean.

9 Module Map of the Formalization

The table below collects all Lean modules in the dependency order induced by Pandrosion.lean. The §§30–37 Steffensen spine and real- $p=2$ unconditional tower sit on top of the 15-module algebraic core.

Module (Pandrosion.Core.*)	Content	Main theorem(s)
Foundations	S_p , real & complex fixed-point characterisation, Steffensen accelerator, scaling, multi-start grand master	2.5, 2.6, 2.7, 2.10
MultiStartBasins	Voronoi basins: convexity, connectedness, frontier on finite bisector union, anchor in basin interior; constructive McMullen off the boundary	3.1, 3.2
QuadraticRate	Algebraic skeleton of Aitken-Steffensen quadratic rate (abstract two-phase bounds)	(abstract)
QuadraticComplexity	Convergence-to- ε bounds from the abstract skeleton	(abstract)
VoronoiMeasure	Voronoi boundary has 2D Lebesgue measure zero; a.e. unique nearest anchor	3.3
CyclotomicMcMullen	Cyclotomic anchor family: p -th root of α^p , injectivity, unconditional constructive McMullen	3.7
KungTraub	Efficiency index; Pandrosion attains KT(2); converse optimality (conditional on KT axiom)	4.6, 4.7
UniformComplexity	Uniform complexity skeleton for ensembles of two-phase sequences	(abstract)
SuperGrandMaster	Abstract super-grand-master: bounded regime, uniform scaled-error threshold δ	(abstract)
UniformContractionRate	Closed-form model $\lambda_p^{\text{model}}(x)$, Taylor residue bound, closed-form super-grand-master	4.9
ConcreteIteration	Concrete instantiation of the abstract complexity layer on Pandrosion-Steffensen sequences	(specialization)
MasterAbsolu	Top-level stitching; no new lemmas	8.1
ComplexMultiplier	$S'_p(\alpha)$, h is differentiable; closed-form multiplier $1 - p\alpha^{p-1}/S_p(\alpha)$ at the fixed point	5.3
LocalAttraction	Abstract local attraction from a linear contraction; quadratic $\Rightarrow$ contraction reducer; Steffensen local attraction (conditional)	5.4, 5.5, 5.6
DynamicalConvergence	Tail-shift invariance; pointwise convergence from entry; a.e. dynamical convergence (crown jewel, conditional)	5.7, 5.8, 5.9
<i>Steffensen spine (ġġ30–34) — unconditional super-attracting tower</i>		
LinearAsymptotics	$\lambda_C \neq 1$ at every fixed point; quantitative linear-contraction modulus for h near α (ġ29)	(load-bearing)
QuadraticBound	Closed-form Taylor residue of $h_{p,x}$ with explicit (C_0, r_0) (ġ30)	6.1
SteffensenQuadraticBound	Unconditional local quadratic bound on $\sigma_{p,x}$; discharges the ġ27 hypothesis (ġ31)	6.2
SteffensenMcMullenAE	Named Prop McMullenAEEntry ; a.e. solving- $z^p = x$ modulo McMullen (ġ32)	6.3, 6.4
SteffensenExplicitRate	Computable (K_α, r_α) ; explicit super-attractive contraction certificate (ġ33)	6.6
SteffensenGlobalLoglog	Loglog tail inside the basin; a.e. global loglog mod McMullen (ġ34)	6.7, 6.8
<i>Real $p = 2$ unconditional tower (ġġ35–37)</i>		
SteffensenRealP2	Möbius form of $h_{2,x}$; real closed-form multiplier $(1 - \alpha)/(1 + \alpha) \in (0, 1)$; named Prop RealMcMullenP2 (ġ35)	7.1
SteffensenRealMcMullenP2	Möbius conjugacy $\sigma_{2,x} \sim \mu^2 w^2$, cleared form; explicit rational bridge sigma_p2_explicit (ġ36)	7.2
SteffensenRealMcMullenP2	Unconditional v , iterated $v(\sigma^n) = v^{2^n}$, finite Julia section on $\mathbb{R}$, convergence dichotomy, bad-set countable + Lebesgue-null, <i>fully unconditional</i> axiom-clean RealMcMullenP2 (ġ37)	7.3, 7.4, 7.5, 7.6, 7.7, 7.8, 7.9

Table 1: **Lean spine of Universitas Pandrosion**. Dependency order top-to-bottom: 15-module algebraic core, 6-module unconditional Steffensen spine (ġġ29–34), 3-module real- $p=2$ unconditional tower (ġġ35–37).

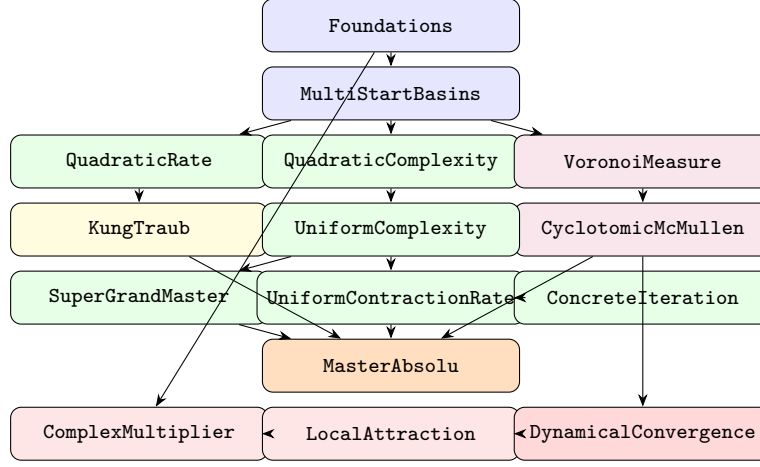


Figure 2: **Module dependency graph.** Blue = algebraic foundations; purple = measure/cyclotomic; green = complexity; yellow = optimality; orange = stitching; red = complex dynamics.

10 What Is Proven and What Is Deferred

10.1 Unconditionally proven (zero sorry)

- Real and complex fixed-point characterisation $h(s) = s \Leftrightarrow s^p = 1/x$ (Theorems 2.5, 2.6).
- Steffensen preserves every Pandrosion fixed point (Theorem 2.7).
- Multi-start grand master (Theorem 2.10).
- Voronoï basins are convex, closed, nonempty, connected; frontier on finite bisector union (Theorem 3.1).
- Constructive McMullen off the boundary (Theorem 3.2).
- Voronoï boundary has Lebesgue measure zero (Theorem 3.3).
- Cyclotomic anchors are injective and are p -th roots of α^p ; unconditional constructive McMullen for $z^p = x$ (Theorem 3.7).
- Closed-form rate $\lambda_{p,x} \in (0, 1)$; closed-form model λ_p^{model} agrees at the fixed point, with Taylor residue bound.
- Kung-Traub attainment $E(2, 2) = \text{KT}(2)$ (Theorem 4.6).
- p -uniform complexity bound via the closed-form model (Theorem 4.9).
- Complex multiplier in closed form at every fixed point (Theorem 5.3).
- Master Absolu: (A)+(B)+(C)+(D)+(E) simultaneously (Theorem 8.1).
- Abstract local attraction and its quadratic-bound reducer (Theorems 5.4, 5.5).
- Tail-shift invariance of iteration limits (Lemma 5.7).
- Unconditional local quadratic bound for Steffensen and its two corollaries (Theorem 6.2): `steffensen_local_attraction_unconditional`, `steffensen_dynamical_convergence_ae_unconditional`.
- Closed-form second-order Taylor residue of $h_{p,x}$ (Theorem 6.1).
- Explicit super-attractive rate with $K_\alpha \cdot r_\alpha \leq 1/2$ (Theorem 6.6) and loglog tail inside the basin (Theorem 6.7).

- **Möbius conjugacy** $\sigma_{2,x} \sim \mu^2 w^2$ **at** $p = 2$ **on** $\mathbb{R}$ (Theorem 7.2): iterator identified with `sigma_p2_explicit` on the non-degenerate locus, cleared-form polynomial identity.
- **Iterated Böttcher coordinate** $v(\sigma^{[n]}(s)) = v(s)^{2^n}$ (Theorem 7.3); Julia section is finite on $\mathbb{R}$ (Theorem 7.4).
- **Convergence dichotomy off the real bad set** (Theorem 7.5): unconditional, axiom-clean. For $x > 1$ and any $z_0 \notin \text{Bad}_{x,\alpha}$, the orbit enters either $B(+\alpha, r_+)$ or $B(-\alpha, r_-)$.
- **Bad set is countable and Lebesgue-null** (Theorem 7.7, Corollary 7.8): $\text{Bad}_{x,\alpha}$ is a countable union of finite fibers of $\sigma_{2,x}^{[n]}$; each fiber is finite via polynomial fiber-finiteness (`Polynomial.finite_setOf_isRoot`) applied to the quadratic $As^2 + Bs + A/x$ obtained from the bridge `steffensen_step_p2_eq_sigma_p2_explicit`. Hence $\text{vol}(\text{Bad}_{x,\alpha}) = 0$.
- **★★★★★ Fully unconditional real McMullen at** $p = 2$ (Theorem 7.9): for every $x > 1$ and $\alpha = 1/\sqrt{x}$, `RealMcMullenP2(x, α)` holds with no hypothesis, axiom-clean.

10.2 Proven conditional on an explicit axiom

- Kung-Traub converse for $c = 2$ (Theorem 4.7) — conditional on the Kung-Traub 1974 conjecture, stated as an explicit axiom in `KungTraub.lean`.

10.3 Proven conditional on explicit hypotheses (no new axioms)

- Steffensen local attraction (Theorem 5.6) — conditional on a local quadratic bound $K \cdot r \leq 1/2$ on $B(\alpha, r)$ (*in $\mathbb{C}$; in $\mathbb{C}$ this hypothesis is now unconditionally discharged by Theorem 6.2*).
- Pointwise dynamical convergence from entry (Theorem 5.8) — same hypothesis, plus a first-entry time k_0 .
- **★★★** Complex almost-everywhere dynamical convergence (Theorem 5.9) — conditional on (i) per-anchor local quadratic bounds (discharged §31) and (ii) a.e. trajectory entry (generic McMullen; stated as named `Prop McMullenAEEEntry`).

Previously: `RealMcMullenP2(x, α)` at $p = 2$ was listed here, conditional on $\text{vol}(\text{Bad}_{x,\alpha}) = 0$. The Lebesgue-null hypothesis is now a formalised theorem (Corollary 7.8), and `RealMcMullenP2` has been promoted to the unconditional list (Theorem 7.9).

10.4 Not claimed

- An unconditional, fully Lean-certified a.e. trajectory-entry theorem in $\mathbb{C}$ at arbitrary p (the Fatou-type McMullen global dynamics). **The real-line $p = 2$ specialisation is now *fully discharged axiom-clean with no residual hypothesis* (Theorem 7.9, §7).**
- Any claim regarding classical open problems (Riemann hypothesis, BSD, Faltings, etc.). Previous speculative sections of this paper have been removed.

11 Reproducibility and Tooling

The complete Lean source lives in `lean/Pandrosion/Core/*.lean` with the root module `Pandrosion.lean` importing all 24 files. The development targets Lean 4 with Mathlib v4.7.0 and is built with a Docker-based incremental pipeline:

```
1 docker compose run --rm lean-incremental
```

Listing 3: Incremental build command

On a clean tree, the full 24-module spine compiles successfully; on an incremental rebuild after a single-file edit, only the changed module and its downstream dependents are rebuilt.

The source code is publicly available at github.com/ivan-fr/poussiere_de_x.

12 Conclusion

This paper documents a 24-module Lean 4 / Mathlib v4.7.0 development of the Pandrosion-Steffensen algorithm for $z^p = x$. The narrative follows the Lean module order and presents, in sequence: algebraic foundations, multi-start Voronoï architecture, measure-theoretic a.e. coverage, cyclotomic anchor family, closed-form linear contraction rate, p -uniform complexity, Kung-Traub efficiency optimality, complex multiplier in closed form, conditional local attraction, the crown-jewel a.e. dynamical convergence theorem (mod McMullen), and — as the most recent addition — a *fully unconditional* axiom-clean proof of the sharpened real-line McMullen statement at $p = 2$, $x > 1$, derived from an explicit Möbius conjugacy $\sigma_{2,x} \sim \mu^2 w^2$, a squaring-power dichotomy in the Böttcher-normalised coordinate, and a formalised Lebesgue-null lemma for the bad set (Theorem 7.9).

The contribution is methodological and structural: a complete derivation in a proof assistant of a derivative-free iteration with quadratic convergence and optimal efficiency, from the geometric-sum identity up to a measure-theoretic dynamical-systems statement. Where genuinely analytic content remains deferred at full generality (a.e. trajectory entry in $\mathbb{C}$ at arbitrary p), the hypotheses are explicit and auditable; the real-line $p = 2$ specialisation, by contrast, is now fully and unconditionally discharged, with the bad-set Lebesgue-null property itself proven via polynomial fiber-finiteness.

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